Lecture 24 – Ampere's Law

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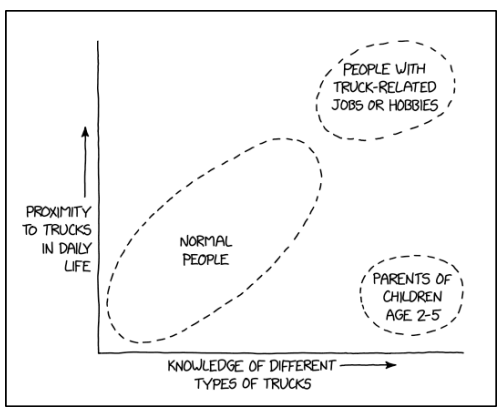

1 Practice with the Lorentz Law

1.1 Task 1

A 2C charge moves with velocity $\vec{v} = 3\hat{i}$ through a $\vec{B} = 4\hat{j}T$ magnetic field. What force does it experience?

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

$$\vec{F} = q * V_x B_y \hat{k} = 2 * 3 * 4 = 24N$$

1.2 Task 2

A magnetic field is magnitude B and points into the board. A particle of charge q is initially travelling speed v to the right.
The particle therefore performs circular motion around it up and down. What is the radius, and is it CW or CCW?

$$F_B = F_{cp}$$

$$qvB = \frac{mv^2}{r}$$

$$r = \frac{mv}{qB}$$

1.3 Task 3

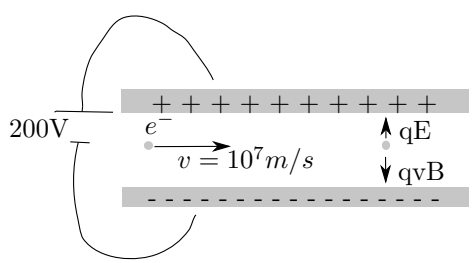

Figure 1: Two really large plates

What magnetic field strength and direction are required to make the electron pass through without being deflected?

Though right-hand rule 2 says the $\vec{B}$ field should be out of the board, since we're dealing with an electron $\vec{B}$ goes into the board. Therefore:

$$qE = qvB$$

$$E = vB$$

$$\frac{V}{d} = vB$$

$$B = \frac{V}{dv}$$

2 Force on a current-carrying wire

$$\vec{F} = I\vec{l} \times \vec{B}$$

$$= l\vec{I} \times \vec{B}$$

In right hand rule 2, $\vec{I}$ is the same magnitude and direction as $\vec{B}$.

2.1 Force between two wires

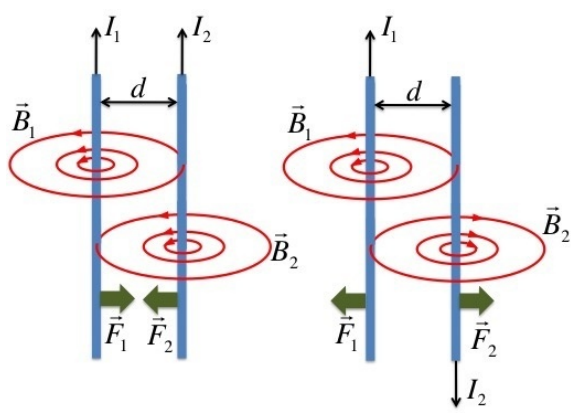

Figure 2: Force between two wires

Wires with current in the same direction attract, while wires with current in different directions will repel.

$$F = I_1 l B_2$$

with

$$B = \frac{M_0 I}{2\pi r}$$

therefore,

$$F = I_1 l B_2 = \frac{M_0 l I_1 I_2}{2\pi r}$$

3 Understanding Ampere's Law

$$\oiint \vec{E}.d\vec{A} = \frac{Q_{in}}{\epsilon_0}$$

$$\oiint \vec{B}.d\vec{A} = 0$$

$$\oint \vec{E}.d\vec{S} = -\frac{d\phi_B}{dt}$$

$$\oint \vec{B}.d\vec{S} = M_0 I_{thru} + M_0 \epsilon_0 \frac{d\phi_E}{dt}$$

The difference between the surface integrals – $\oiint$ is a two-dimensional surface integral over gaussian surfaces, while $\oint$ is a one-dimensional line integral over closed Amperean loops.

Analyzing the last surface integral:

$$\oint \vec{B}.d\vec{S} = M_0 I_{thru} + M_0 \epsilon_0 \frac{d\phi_E}{dt}$$

- $d\vec{S}$ is a line element.
- M_0 is the magnetic constant, permeability of free space.
- I_{thru} is the current through the Amperean loop.

3.1 What is $\int \vec{B}.d\vec{S}$?

A:

$$\int_i^f \vec{B}.d\vec{S} = \int_i^f B ds = BS$$

B:

$$\int_i^f \vec{B}.d\vec{S} = 0$$

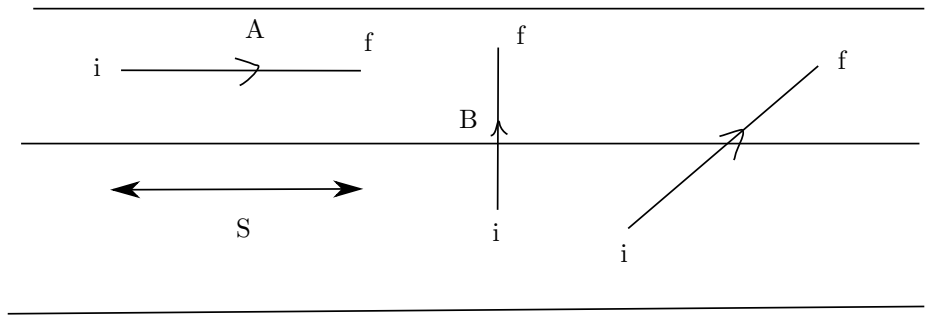

Figure 3: BdS diagram

C:

$$\int_i^f \vec{B}.d\vec{S} = \int_i^f BdS\cos\theta$$